Tuple:

1. Tuple is similar to List except that the objects in tuple are immutable which means we cannot change the elements of a tuple once assigned.

2. When we do not want to change the data over time, tuple is a preferred data type

In [1]:
```
t1 = ()    #empty tuple
t1
```

Out[1]:
```
()
```

In [2]:
```
t2 = tuple()   #empty tuple
t2
```

Out[2]:
```
()
```

In [3]:
```
t3=(2,3,True,2.23,[2,3,5],{10,20,30}) #tuple is heterogenous
t3
```

Out[3]:
```
(2, 3, True, 2.23, [2, 3, 5], {10, 20, 30})
```

In [4]:
```
len(t3) #length of tuple
```

Out[4]:
```
6
```

In [5]:
```
t4 = (2, 3, True, 2.23, [2, 3, 5], (10, 20, 30)) #nested tuple
t4
```

Out[5]:
```
(2, 3, True, 2.23, [2, 3, 5], (10, 20, 30))
```

slicing in tuple

In [6]:
```
t4
```

Out[6]:
```
(2, 3, True, 2.23, [2, 3, 5], (10, 20, 30))
```

In [7]:
```
t4[-1]
```

Out[7]: (10, 20, 30)

In [8]:
```
t4[-1][1]
```

Out[8]: 20

In [9]:
```
t4
```

Out[9]: (2, 3, True, 2.23, [2, 3, 5], (10, 20, 30))

In [10]:
```
t4[1:6]
```

Out[10]: (3, True, 2.23, [2, 3, 5], (10, 20, 30))

In [11]:
```
t4
```

Out[11]: (2, 3, True, 2.23, [2, 3, 5], (10, 20, 30))

In [12]:
```
t4[-1]
```

Out[12]: (10, 20, 30)

In [13]:
```
t4[-1][0] #nested indexing
```

Out[13]: 10

In [14]:
```
t5 = (5,8,9,5,6,20,80,55,25,25,'true', 6)
t5
```

Out[14]: (5, 8, 9, 5, 6, 20, 80, 55, 25, 25, 'true', 6)

In [15]:
```
t5.count(25)
```

Out[15]: 2

In [16]:
```
t5.count('true')
```

Out[16]: 1

In [17]:
```
t5[::-1] #bavkward slicing
```

Out[17]: (6, 'true', 25, 25, 55, 80, 20, 6, 5, 9, 8, 5)

In [18]:
```
t5
```

Out[18]: (5, 8, 9, 5, 6, 20, 80, 55, 25, 25, 'true', 6)

In [19]:
```
t5.index('true')
```

Out[19]: 10

Remove and change iteam in tuple

In [20]:
```
t5
```

Out[20]:
```
(5, 8, 9, 5, 6, 20, 80, 55, 25, 25, 'true', 6)
```

In [21]:
```
t5[0]=100 # Tuples are immutable which means we can't CHANGE tuple items
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[21], line 1
----> 1 t5[0]=100 # Tuples are immutable which means we can't CHANGE tuple items

TypeError: 'tuple' object does not support item assignment
```

In [ ]:
```
t5
```

In [ ]:
```
t3
```

In [ ]:
```
del t3[1] # Tuples are immutable which means we can't DELETE tuple items
```

In [ ]:
```
del t3 # Deleting entire tuple object is possible
```

sorting in tuple

In [ ]:
```
tup = (100,200,300,400,500,1000)
tup
```

In [ ]:
```
sorted(tup)  #bydefault in increasing order
```

In [ ]:
```
sorted(tup, reverse = True) #decending order
```

loop in tuple

In [ ]:
```
tup
```

In [ ]:
```
for i in tup :
    print(i)
```

In [ ]:
```
for i in enumerate(tup):
    print(i)
```

convert tuple to list

In [ ]:
```
tup
```

In [ ]:
```
l = list(tup)
l
```

In [ ]:
```
print(tup)
print(l)
```

In [ ]:
```
id(l),id(tup)
```

if we want to modify iteam in tuple then:

1.convert tuple to list

2.modify iteam

3.convert list to tuple

In [ ]:
```
tup
```

In [ ]:
```
l = list(tup)
l.append(25)
print(l)
```

In [ ]:
```
tup2 = tuple(l)
print(tup2)
print(l)
```

In [ ]:

In [ ]: